



Combined Delaware Basin Stage 5 Project and Alternative ERCOT Independent Review Status Update

Tanzila Ahmed

RPG Meeting
January 28, 2025

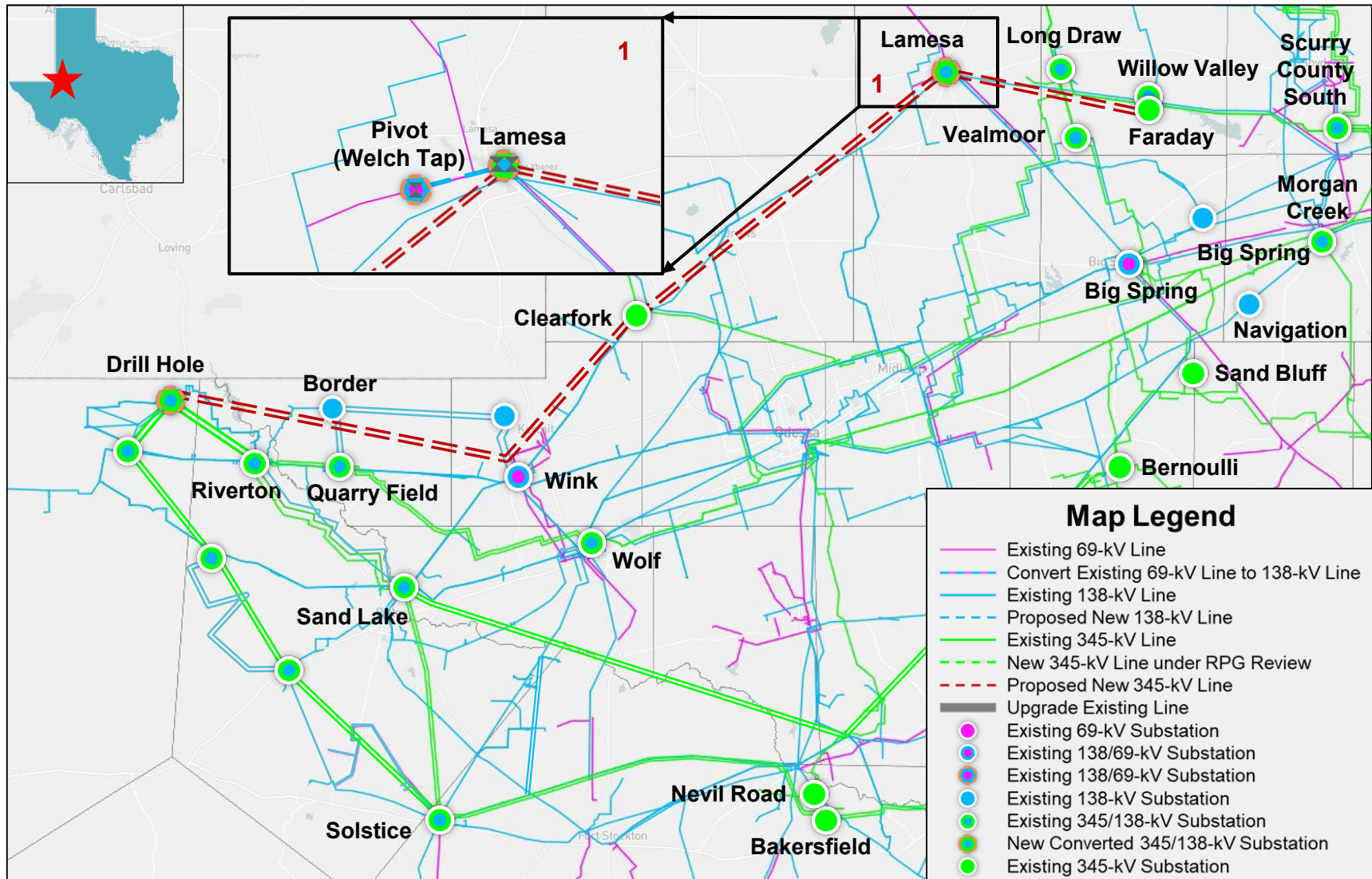
Recap Introduction

- Oncor Electric Delivery (Oncor) submitted the Delaware Basin Stage 5 for Regional Planning Group (RPG) review in May 2024
 - This Tier 1 project is estimated to cost \$744.6 million
 - Filing of Certificate of Convenience and Necessity (CCN) will be required
 - Estimated in-service date (ISD) is December 2029
 - The project is needed to address reliability issues (both thermal overloads and voltage violations) due to significant load growth in the Delaware Basin area, in the Andrews, Borden, Culberson, Dawson, Gaines, Loving, Reeves, and Winkler Counties in the Far West (FW) Weather Zone
 - The project need and solution was identified in the 2019 ERCOT Delaware Basin Load Integration Study
 - Background recapped in [Appendix C](#)
- Wind Energy Transmission Texas, LLC (WETT) submitted an alternative option for RPG review in June 2024
 - WETT's portion of this option is estimated to cost \$305.5 million
 - WETT estimates a total cost saving of up to approximately \$67.0 million
 - Filing of Certificate of Convenience and Necessity (CCN) will be required
 - Estimated in-service date is December 2028 for WETT's portion of the project

Recap Introduction (cont.)

- ERCOT is currently conducting a single ERCOT Independent Review (EIR) by combining these two projects (24RPG015 and 24RPG023)
- Oncor and WETT presented project overviews and ERCOT presented the scope for this EIR at the September RPG meeting
 - <https://www.ercot.com/calendar/09252024-RPG-Meeting>

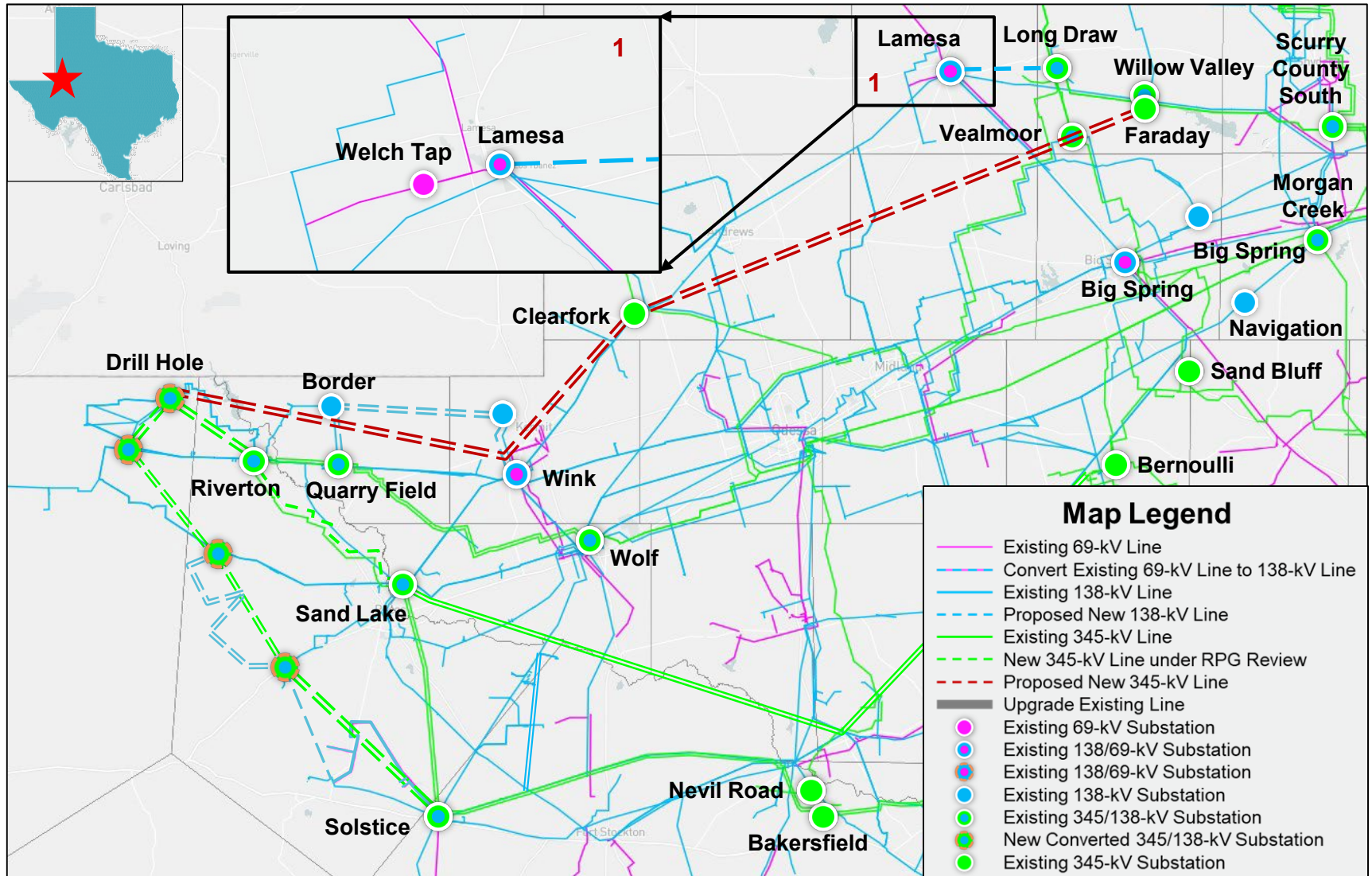
Recap Map – Proposed Project by Oncor



Recap Proposed Project by Oncor

- Expand the existing Lamesa Switch, including a 13-breaker 138-kV breaker-and-a-half bus arrangement and a 9-breaker 345-kV breaker-and-a-half bus arrangement with two 600 MVA, 345/138-kV autotransformers. All terminal and associated equipment will meet or exceed 5000 A for 345-kV and 3200 A for 138-kV
- Construct a new Clearfork Switch – Lamesa Switch 345-kV double-circuit transmission line using a normal and emergency rating of at least 2988 MVA, which will require new right of way (ROW), approximately 77.0-mile
- Construct a new Lamesa Switch – Faraday Switch 345-kV double-circuit transmission line using a normal and emergency rating of at least 2988 MVA, which will require new ROW, approximately 38.0-mile
- Expand the existing Faraday 345-kV substation to accommodate the new Lamesa Switch – Faraday Switch 345-kV double-circuit transmission line and loop in the existing Long Draw – Scurry 345-kV transmission line
- Establish a new Pivot 138/69-kV Switch in the current Welch Tap location which will include
 - Relocating one of the existing Lamesa 138/69-kV autotransformers and three of the existing Lamesa 69-kV breakers to the new Pivot 138/69-kV Switch
 - Rebuilding and converting the existing Lamesa Switch – Welch Tap 69-kV transmission line to 138-kV operation using a normal and emergency rating of at least 614 MVA, 2.0-mile
- Rebuild the existing Clearfork 345-kV Switch by installing thirteen 5000 A, 345-kV circuit breakers in a breaker-and-a-half bus arrangement
- Install two 5000 A, 345-kV circuit breakers in a breaker-and-a-half bus arrangement at the planned Drill Hole 345-kV Switch
- Construct a new Clearfork Switch – Drill Hole Switch 345-kV double-circuit transmission line using a normal and emergency rating of at least 2988 MVA, which will require new ROW, approximately 105.0-mile
 - This line will be routed near the location of planned Border Switch for a future 345-kV interconnection to provide an injection point to support this high load growth area

Recap Map – Proposed Project by WETT



Recap Proposed Project by WETT

- Construct a new Faraday Switch – Clearfork Switch 345-kV double-circuit line using a normal and emergency rating of at least 2988 MVA, which will require new ROW, approximately 105.0-mile
- Expand the existing Faraday 345-kV substation to accommodate the new Faraday Switch - Clearfork Switch 345-kV double-circuit transmission line and loop in the existing Long Draw – Scurry 345-kV transmission line
- Construct a new Long Draw Switch – Lamesa Switch 138-kV transmission line using a normal and emergency rating of at least 614 MVA, which will require new ROW, approximately 22.0-mile
- Expansion of the existing Long Draw 138-kV substation and the existing Lamesa 138-kV substation to accommodate the new Long Draw – Lamesa 138-kV transmission line
- Rebuild the existing Clearfork 345-kV Switch by installing thirteen 5000 A, 345-kV circuit breakers in a breaker-and-a-half bus arrangement
- Install two 5000 A, 345-kV circuit breakers in a breaker-and-a-half bus arrangement at the planned Drill Hole 345-kV Switch
- Construct a new Clearfork Switch – Drill Hole Switch 345-kV double-circuit transmission line using a normal and emergency rating of at least 2988 MVA, which will require new ROW, approximately 105.0-mile
 - This line will be routed near the location of planned Border Switch for a future 345-kV interconnection to provide an injection point to support this high load growth area

Recap Study Assumptions

- Final 2023 Regional Transmission Planning (RTP) 2029 summer peak case for West and Far West (WFW) Weather Zones was used as the starting case
- Transmission Updates
 - New transmission projects, listed in the [Appendix A1](#), based on the June 2024 Transmission Project and Information Tracking (TPIT) report along with recently approved RPG projects were added to the base case
 - A placeholder project, listed in [Appendix A2](#), for the Delaware Basin Stage 5 upgrade was removed from the base case
 - Among the local transmission upgrades in the Permian Basin Reliability Plan Study for the Delaware Basin region, upgrades ID L1, L3, L5, and L6 that were added as part of the study base case for the Delaware Basin Stages 3 and 4 Project, were added to the study base case. The components for these local upgrades are listed in [Appendix A3](#)

Recap Study Assumptions (Continued)

- Generation update
 - New 6.9(1) generation, listed in [Appendix B](#), were added based on the August 2024 Generator Interconnection Status (GIS) report
 - All generation were dispatched consistent with the 2024 RTP methodology
 - Reserve was maintained consistent with 2024 RTP methodology
- Loads update
 - Oil & Gas loads in the FW Weather Zone were updated based the S&P Global Load Forecast
 - Large Loads with Signed Interconnection Agreement (IA) were added in the FW Weather Zone

Load Type	Load (~MW)
Total S&P Global Forecasted Loads	8,905
Total New Large Loads Added	1,194
Total FW Weather Zone Loads	16,389

Status Update

- Project Evaluation
 - Reliability Analysis
 - N-1, G-1+N-1, and X-1+N-1
 - G-1: Permian Basin all five units, and Odessa Combined Cycle (CC) train 1
 - X-1: Drill Hole, Riverton, Long Draw, and Willow Valley 345/138-kV transformers
 - Long-Term Load-Serving Capability Evaluation

Preliminary Results of Reliability Assessment – Options Evaluation

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC TPL-001-5.1 and ERCOT Planning Criteria to evaluate the proposed options

Option	N-1		G-1* + N-1		X-1** + N-1		Unsolved Powerflow
	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	
ONCOR	None	None	None	None	None	None	None
WETT	None	None	None	4	None	None	None

** G-1: Odessa Ector CC Train, and Permian Basin all five units

*** X-1: Drill Hole, Riverton, Long Draw and Willow Valley 345/138-kV transformers

Long-Term Load-Serving Capability Evaluation

- Assumptions
 - Adjusted load up in the Lamesa area, excluding Flexible Loads in the area
 - Adjusted conforming load down outside of Far West to balance power
 - Based on N-1 contingency
- Preliminary Findings

Option	Incremental Load Serving Capability into Lamesa area (~MW)
Oncor	365
WETT	87

Next Steps

- ERCOT will continue with project evaluation and perform the following
 - Conduct Planned Maintenance Outage Evaluation
 - Transmission Service Providers (TSPs) to conduct Cost Estimate and Feasibility Assessment
- Additional analyses may be performed on the preferred option
 - Congestion Analysis to ensure that the identified transmission upgrades do not result in new congestion within the study area
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
 - Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)

Deliverables

- Tentative Timelines
 - Status updates at future RPG meetings
 - Final recommendation – Q1 2025

Thank you!



Stakeholder comments also welcomed through:

Tanzila.Ahmed@ercot.com

Robert.Golen@ercot.com

Appendix A1 – Transmission Projects Added

TPIT No	Project Name	Tier	Project ISD	County
6719	Twelvemile Substation Addition	4	5/30/2025	Pecos
72863	Delaware River 138-kV Switch	4	5/15/2024	Culberson
72935	Saragosa: Install 2nd Bay	4	9/30/2024	Reeves
73381	TNMP_JACKRABBIT_CUTIN_AC_4-5-2023	4	8/2/2023	Pecos
73452	TNMP_WINK_FISHHOOK_RECONDUCTOR_AC_4-5-2023	4	1/31/2024	Pecos
73476	TNMP_KERMIT_RECONDUCTOR	4	12/31/2024	Pecos
76151	Gas Pad Tap: Replace CTVT	4	4/30/2024	Reeves
76174	Origin 138-kV Interconnection	4	6/30/2025	Reeves
76212	Model Coachwhip Sub	4	5/31/2024	Ward
76232	Reconductor Mivida-Coachwhip-Fishhook 2045 ACCC	4	5/31/2026	Ward
76291	Upgraded Cedarvale–BoneSpringsTap–Fishhook	4	5/31/2026	Ward
76293	Upgraded Cedvale-MiDiva138KV	4	5/31/2026	Ward
76348	Reconductor Foxtail-PIGCreek-1926ACSS-138KV	4	5/31/2026	Pecos
76696	Construct a new Border – Shifting Sands 138-kV Line	2	12/15/2026	Loving
76719	Establish Bull Moose 138-kV Switch	4	12/15/2024	Loving
77146	Reconductor WNK-AAT-MDT-FSH	4	1/31/2024	Winkler
77320	Add CapBANK in COYANOSA	4	6/1/2026	Ward
78044	ROCK DRAW 345-kV Switch	3	12/30/2026	Ward
78046	TOYAH CREEK 345-kV Switch	3	12/30/2026	Ward

Appendix A1 – Transmission Projects Added (cont.)

TPIT No	Project Name	Tier	Project ISD	County
22RPG045	Yucca to Moss 138-kV Line Project	4	5/1/2024	Ward
23RPG013	Silverleaf and Cowpen 345/138-kV Stations Project	1	5/31/2025	Ward, Reeves
23RPG023	Pecos County Transmission Improvement Project	1	6/1/2027	Pecos
23RPG027	Bakersfield Dynamic Reactive Substation Upgrade	1	8/31/2026	Pecos
24RPG002	Rockhound 345/138-kV Switch and Grey Well Draw to Buffalo 2nd 138-kV Circuit Project	3	12/24/2024	Martin
23RPG034	West Texas 345-kV Infrastructure Rebuild Project	1	6/15/2028	Scurry, Mitchell, Howard, Glasscock, Martin, Midland, Ector
24RPG010	Delaware Basin Stages 3 and 4 Project	1	Summer 2027	Culberson, Loving, Reeves, Ward
24RPG027	Long Draw Second 345/138-kV Autotransformer Project	3	5/1/2026	Borden

Appendix A2 – 2023 RTP Placeholder Projects Removed

RTP Project ID	Project Name	County(s)
2022-FW24	Faraday - Lamesa - Clearfork - Riverton 345-kV Double Circuit Line Addition	Borden, Dawson, Andrews, Winkler, Loving and Reeves

Appendix A3 – Local Upgrades from 2024 Permian Basin Reliability Plan Study Added

Project No	Project Name
L1	Add Quarry Field – Border 138-kV second circuit
L1	Add Wink – Shifting Sands 138-kV second circuit
L1	Connect new load bus 900004 to Border and new load bus 900066 to Shifting Sands to form a 138-kV double-circuit loop
L1	Add Riverton – Border 138-kV second circuit
L3	Connect new load buses to 11610 and Faulkner and form a 138-kV double-circuit loop: 11610 – 900005 – 900111 – 900023 – 900012 – 900021 – 900038 – 38124
L4	Reconnect new load 900108 to Cryo instead of Saragosa
L5	Establish a new Culberson 345/138-kV substation at the existing Culberson Switch and install two new 345/138-kV transformers
L5	Establish a new ONC900005_TAP 345/138-kV substation and install two new 345/138-kV transformers
L5	Establish a new Faulkner 345/138-kV substation at Faulkner station and install two new 345/138-kV transformers
L5	Add a new Drill Hole – Culberson 345-kV double-circuit line
L5	Add a new Culberson – 900005Tap 345-kV double-circuit line
L5	Add a new 900005Tap – Faulkner 345-kV double-circuit line
L5	Add a new Solstice – Faulkner 345-kV double-circuit line
L6	Upgrade Existing Cowpen - Birds of Pray Tap 138-kV Line N-1-1

Appendix B – New Generation Added

GINR	Project Name	Fuel	Projected COD	Capacity (~MW)	County
19INR0203	Angelo Solar	SOL	8/1/2024	195.4	Tom Green
21INR0424	Tierra Bonita Solar	SOL	9/26/2024	306.9	Pecos
22INR0502	Shamrock Wind SLF	WIN	4/19/2024	223.9	Crockett
23INR0219	Dogfish BESS	OTH	4/16/2025	77.4	Pecos
23INR0372	Cross Trails Storage	OTH	4/25/2025	58.3	Scurry
23INR0387	Pioneer DJ Wind	WIN	9/15/2024	140.3	Midland
23INR0401	Headcamp BESS	OTH	6/16/2025	152.9	Pecos
23INR0418	Angelo Storage	OTH	8/10/2024	103.0	Tom Green
23INR0470	BoCo BESS	OTH	7/17/2024	155.5	Borden
24INR0273	Al Pastor BESS	OTH	8/30/2024	103.1	Dawson
24INR0421	Swift Air Solar	SOL	2/12/2025	148.8	Ector
24INR0578	Panther Creek 1 Repower	WIN	10/30/2024	11.0	Glasscock
24INR0582	Panther Creek 2 Repower	WIN	10/30/2024	8.0	Glasscock
24INR0627	Champion Wind Repower	WIN	5/9/2025	0.3	Nolan
25INR0400	Maldives Solar (Alternate POI)	SOL	7/1/2027	210.0	Scurry

Appendix C – Background

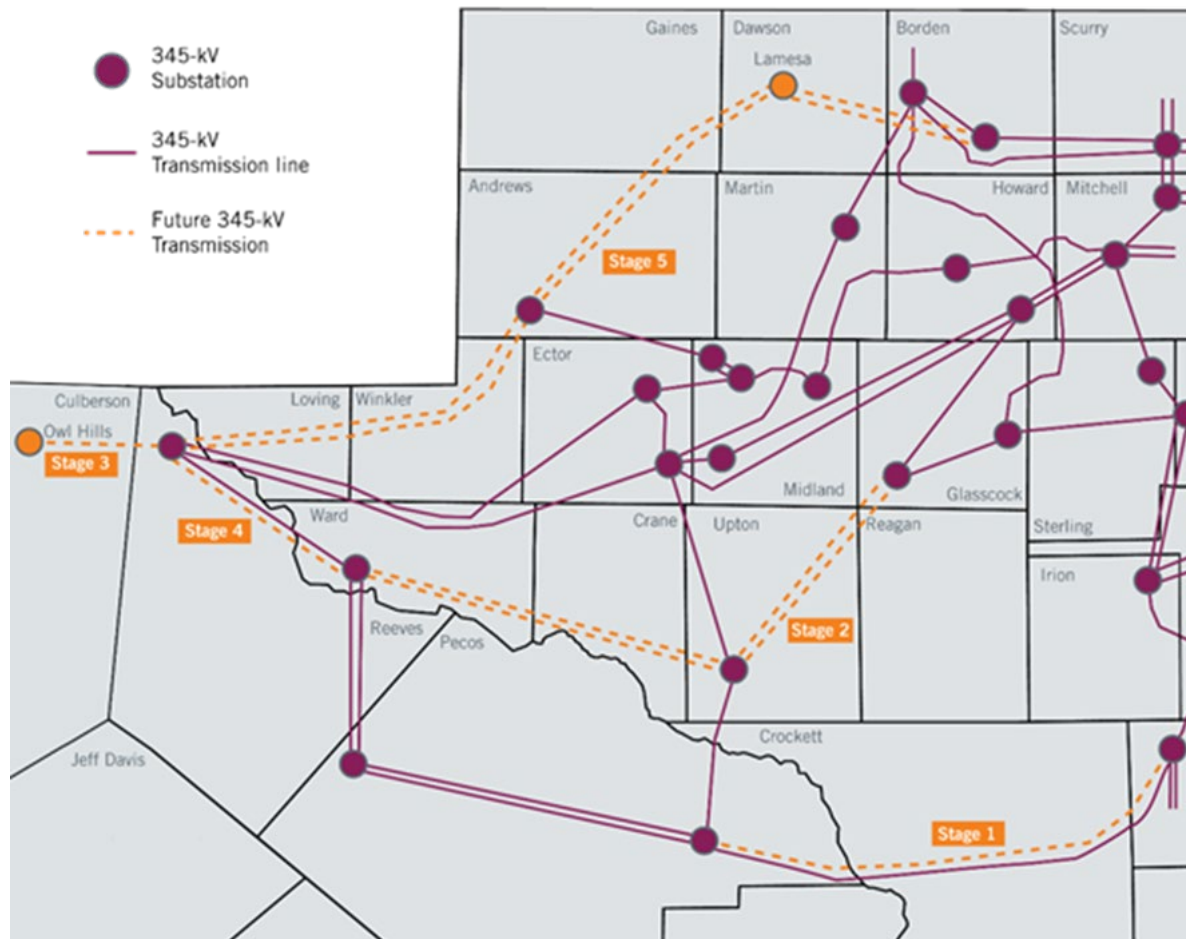
- The Far West Weather Zone, which includes the Study area for this project, has experienced significant growth in oil and natural gas industry demand
- Due to the significant load growth and because of lack of long-term load commitment from the oil and gas customers, ensuring that necessary transmission improvements are in place in time is a significant challenge for both ERCOT and TSPs
- As part of the efforts to address this challenges, ERCOT previously completed two studies which incorporated extensive review and input by Transmission Service Providers (TSPs) and stakeholders:
 - 2019 Delaware Basin Load Integration Study
 - 2024 Permian Basin Reliability Plan Study

Appendix C – Background (Cont.)

- [Delaware Basin Load Integration Study](#) completed in December 2019
 - Identified the reliability needs in the region
 - Provided a roadmap of long lead time transmission improvements for the continued oil and gas load growth in the Delaware Basin area
 - Stage 1 upgrade was endorsed in June 2021 and is expected to be complete in 2023
 - Stage 2 upgrade was endorsed in August 2022 and is expected to be complete in 2026
 - Updated Stage 3 upgrade along with Stage 4 upgrade are currently under RPG review and has an expected ISD of Summer 2027

Stage	Estimated Delaware Basin Load Level (MW)	Upgrade Element	Trigger
1	3,052	Add a second circuit on the existing Big Hill - Bakersfield 345-kV line	Import Needs
2	4,022	A new Bearkat - North McCamey - Sand Lake double-circuit 345-kV line	Import Needs
3	4,582	A new Riverton - Owl Hills single-circuit 345-kV line	Culberson Loop Needs
4	5,032	Riverton - Sand Lake 138-kV to 345-kV conversion and a new Riverton - Sand Lake 138-kV line	Culberson Loop Needs
5	5,422	A new Faraday - Lamesa - Clearfork - Riverton double-circuit 345-kV line	Import Needs

Appendix C – Map of Preferred 2019 Delaware Basin Load Integration Study Upgrades



Appendix C – Background (Cont.)

- [Permian Basin Reliability Plan Study](#) completed in July 2024
 - Identified the reliability needs in the region
 - Identified local upgrades along with import paths needed to serve the Permian Basin region loads
 - Updated Stages 3 and 5 upgrades as well as the Stage 4 upgrade from 2019 Delaware Basin Load Integration Study was included in the base case